

如何使用

在 code 中 build 文件下使用 make 命令编译，编译成功后，在 build 文件夹下会生成可执行文件 P, 在 test 目录下的 test.josn 调整初始条件即可使用不同的测试用例。

2 维情形下运行较慢，因为在插值时对向量进行了多次复制，导致效率较低。

Problem $\exp(y + \sin(x))$

1 维 VCycle

规则边界，Dirichlet 边界条件，fullWeighting 松弛，linear 插值。

1. 网格数为 32。在第 6 次迭代时残差达到 10^{-8}

Residual norm at level 32: 4.37168e-06

Residual norm at level 16: 6.96165e-07

Residual norm at level 8: 8.58941e-08

6 iteration, error = 0.000467973

2. 网格数为 64。在第 6 次迭代时残差达到 10^{-8}

Residual norm at level 64: 1.97495e-05

Residual norm at level 32: 3.03391e-06

Residual norm at level 16: 5.67177e-07

Residual norm at level 8: 6.33308e-08

6 iteration, error = 0.000165485

3. 网格数为 128。在第 6 次迭代时残差达到 10^{-8}

Residual norm at level 128: 8.39476e-05

Residual norm at level 64: 1.29986e-05

Residual norm at level 32: 2.38244e-06

Residual norm at level 16: 4.67866e-07

Residual norm at level 8: 3.95246e-08

6 iteration, error = 5.85109e-05

4. 网格数为 256。在第 5 次迭代时残差达到 10^{-6}

Residual norm at level 256: 0.0162303

Residual norm at level 128: 0.0026792

Residual norm at level 64: 0.000481704

Residual norm at level 32: 7.97636e-05

Residual norm at level 16: 1.11288e-05

Residual norm at level 8: 2.60593e-06

5 iteration, error = 2.06719e-05

规则边界, Dirichlet 边界条件, fullWeighting 松弛, quadratic 插值。

网格数为 256。在第 6 次迭代时残差达到 10^{-4}

Residual norm at level 256: 3.23331

Residual norm at level 128: 0.66233

Residual norm at level 64: 0.119562

Residual norm at level 32: 0.0195825

Residual norm at level 16: 0.00293039

Residual norm at level 8: 0.000371142

6 iteration, error = 8.5186e-05

说明二次插值的精度更高。

规则边界, Dirichlet 边界条件, injection 松弛, linear 插值。在第 6 次迭代时残差达到 10^{-5}

Residual norm at level 256: 0.0430957

Residual norm at level 128: 0.0116944

Residual norm at level 64: 0.00297286

Residual norm at level 32: 0.000708022

Residual norm at level 16: 0.000153124

Residual norm at level 8: 2.5387e-05

6 iteration, error = 2.15675e-05

说明 fullweighting 松弛的精度更高。

1 维 FMGCycle

规则边界, Dirichlet 边界条件, fullWeighting 松弛, linear 插值。不知为何 FMGCyle 方法的误差不会收敛到 0 而是缓慢增大。

1. 网格数为 32。

Residual norm at level 8: 1.12119

Residual norm at level 16: 1.40655

Residual norm at level 8: 0.121142

Residual norm at level 32: 1.36816

Residual norm at level 16: 0.0915754

Residual norm at level 8: 0.0363852

1 iteration, error = 0.000407491

2. 网格数为 64。

Residual norm at level 8: 1.12107

Residual norm at level 16: 1.40635

Residual norm at level 8: 0.121133

Residual norm at level 32: 1.36792

Residual norm at level 16: 0.0915584

Residual norm at level 8: 0.0363813

Residual norm at level 64: 1.35378

Residual norm at level 32: 0.0850067

Residual norm at level 16: 0.0288001

Residual norm at level 8: 0.00902197

1 iteration, error = 0.000170015

在第 5 次迭代时误差增加到, 5 iteration, error = 0.000260534??

3. 网格数为 128。

Residual norm at level 8: 1.12104

Residual norm at level 16: 1.4063

Residual norm at level 8: 0.12113

Residual norm at level 32: 1.36786

Residual norm at level 16: 0.0915542

Residual norm at level 8: 0.0363804

Residual norm at level 64: 1.35371

Residual norm at level 32: 0.0850023

Residual norm at level 16: 0.0287987

Residual norm at level 8: 0.00902171

Residual norm at level 128: 1.36373

Residual norm at level 64: 0.0822714

Residual norm at level 32: 0.0268418

Residual norm at level 16: 0.00725061

Residual norm at level 8: 0.00221144

1 iteration, error = 6.30315e-05

在第 5 次迭代时误差增加到, 5 iteration, error = 9.38406e-05 ??

4. 网格数为 256。Residual norm at level 8: 1.12103

Residual norm at level 16: 1.40629

Residual norm at level 8: 0.12113

Residual norm at level 32: 1.36784

Residual norm at level 16: 0.0915531

Residual norm at level 8: 0.0363801

Residual norm at level 64: 1.3537

Residual norm at level 32: 0.0850012

Residual norm at level 16: 0.0287984

Residual norm at level 8: 0.00902165

Residual norm at level 128: 1.36371

Residual norm at level 64: 0.0822703

Residual norm at level 32: 0.0268414

Residual norm at level 16: 0.00725052

Residual norm at level 8: 0.00221143

Residual norm at level 256: 1.40542

Residual norm at level 64: 0.0262803

Residual norm at level 32: 0.00675963

Residual norm at level 16: 0.00176793

Residual norm at level 8: 0.000548963

1 iteration, error = 2.24901e-05

在第 5 次迭代时误差增加到, 5 iteration, error = 3.32245e-05.??

剩下的松弛方法和插值方法不再一一展示。

2 维 VCycle

二维情形误差较大的原因可能是矩阵组装出现失误, 暂时没有找到原因。

在 Multigrid.h 文件第八百行, 我调库算出所欲求解的方程组的解 (在方程组维数为 8 时), 得到的结果确实为我多重网格的结果, 图片贴在下面, 说明确实是在组装矩阵时出现了问题。

```
-0.029427 -0.0559533 -0.0675724 -0.0669486 -0.0554957 -0.0339366 -0.00226776 -0.0598593 -0.108472 -0.131629 -0.133052 -0.115057 -0.0789899 -0.0253302 -0.0793343
-0.142986 -0.174027 -0.176748 -0.154059 -0.107771 -0.0387966 -0.0893313 -0.160866 -0.195572 -0.19846 -0.172888 -0.120866 -0.0433126 -0.0886138 -0.159773 -0.193214 -0.194929 -0.168564 -0.11601 -0.0376417 -0.0730443 -0.133165 -0.159444 -0.158621 -0.142986 -0.0876961 -0.0185011 -0.0337901 -0.0687462 -0.0812323 -0.076929
-0.058323 -0.0262163 0.0204471 Residual norm at level 8: 6.7462e-07
-0.0285492 -0.0497912 -0.00808166 -0.0568244 -0.0891664 -0.033889 -0.0207657 -0.0423239 0.00565063 6 iteration, error = sol =
```

```
sol =
1 -0.029427
2 -0.0559533
3 -0.0675724
4 -0.0669486
5 -0.0554957
6 -0.0339366
7 -0.00226776
8 -0.0598593
9 -0.108472
10 -0.131629
11 -0.133052
12 -0.115057
13 -0.0789899
14 -0.0253302
15 -0.0793343
16 -0.142986
17 -0.174027
18 -0.176748
19 -0.154059
20 -0.107771
21 -0.0387966
22 -0.0893313
23 -0.160866
24 -0.195572
25 -0.19846
26 -0.172888
27 -0.120866
28 -0.0433126
29 -0.0886138
30 -0.159773
31 -0.193214
32 -0.194929
33 -0.168564
34 -0.11601
35 -0.0376417
36 -0.0730443
37 -0.133165
38 -0.159444
39 -0.158621
40 -0.142986
41 -0.0876961
42 -0.0185011
43 -0.0337901
44 -0.0687462
45 -0.0812323
46 -0.076929
```

因为我在做线性插值的时候为了方便, 建立了一个细网格数量平方的容器用来装插值过后得到的解向量, 导致运行很慢, 256 基本无法运行。

规则边界, Dirichlet 边界条件, fullWeighting 松弛, linear 插值。

1. 网格数为 32。在第 6 次迭代时残差达到 10^{-5}

Residual norm at level 32: 0.000421504

Residual norm at level 16: 7.55235e-05

Residual norm at level 8: 9.04606e-06

6 iteration, error = 35.8642

2. 网格数为 64。在第 6 次迭代时残差达到 10^{-5} Residual norm at level 64: 0.00344576

Residual norm at level 32: 0.000405106

Residual norm at level 16: 8.10676e-05

Residual norm at level 8: 1.00161e-05

6 iteration, error = 71.6858

3. 网格数为 128。在第 6 次迭代时残差达到 10^{-5} RResidual norm at level 128: 0.020186 Residual norm at level 64: 0.00267631 Residual norm at level 32: 0.000334808 Residual norm at level 16: 7.76124e-05 Residual norm at level 8: 1.03135e-05 6 iteration, error = 143.299

2 维 FMGCycle

规则边界, Dirichlet 边界条件, fullWeighting 松弛, linear 插值。

1. 网格数为 32。在第 11 次迭代时残差达到 10^{-3}

Residual norm at level 8: 0.0048538

Residual norm at level 16: 0.0473602

Residual norm at level 8: 0.00300494

Residual norm at level 32: 0.194175

Residual norm at level 16: 0.0347955

Residual norm at level 8: 0.00288642

11 iteration, error = 32.054

2. 网格数为 64。在第 11 次迭代时残差达到 10^{-3}

Residual norm at level 8: 0.010503

Residual norm at level 16: 0.109764

Residual norm at level 8: 0.00677214

Residual norm at level 32: 0.450197

Residual norm at level 16: 0.0830837

Residual norm at level 8: 0.00642242

Residual norm at level 64: 1.83493

Residual norm at level 32: 0.34701

Residual norm at level 16: 0.0788803

Residual norm at level 8: 0.00615708

11 iteration, error = 64.0267

3. 网格数为 128。在第 11 次迭代时残差达到 10^{-3}

Residual norm at level 8: 0.00988042

Residual norm at level 16: 0.135318

Residual norm at level 8: 0.00750912

Residual norm at level 32: 0.587086

Residual norm at level 16: 0.10339

Residual norm at level 8: 0.00694519

Residual norm at level 64: 2.38983

Residual norm at level 32: 0.466767

Residual norm at level 16: 0.0979731

Residual norm at level 8: 0.00635629

Residual norm at level 128: 10.1725

Residual norm at level 64: 1.7243

Residual norm at level 32: 0.44599

Residual norm at level 16: 0.0945225

Residual norm at level 8: 0.00566489

11 iteration, error = 128.013